

Scalable Optimization of Social Welfare

A Parametric Family of Additive Scoring Rules Grounded in Welfare Economics

Olivier Rocca

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Abstract

This article introduces the WO (Welfare Optimization) family of rules, a general framework for collective decision-making based on parametric additive scoring rules.

The central idea is straightforward: instead of using a fixed voting rule (such as plurality or approval), we propose a family of rules capable of explicitly adapting to different social welfare objectives.

Our main result is a complete representation theorem (Theorem 4.1) showing that the WO family corresponds exactly to the set of additive rules that satisfy two fundamental properties:

- the score increases as the approval level increases,
- the score decreases as rejection increases.

This general framework makes it possible to:

- reproduce the major families of welfare economics criteria (utilitarianism, egalitarianism, elitism, Nash),
- rigorously analyze the properties of voting rules,
- and compare their strategic behavior.

We also establish:

- several essential robustness and scalability properties,
- a complete analysis of strategic behavior (including the Gibbard–Satterthwaite impossibility),
- and a classification of rules using a nine-archetype taxonomy, constructed via a reproducible fuzzy logic method.

Finally, we clearly identify the limits of the WO framework, by showing which classes of rules cannot be included within it.

Keywords: *welfare optimization, additive scoring rules, Young's representation theorem, scalable voting, support vector, approval vector, archetype taxonomy, strategic manipulation, interpersonal utility, distributed computation.*

1. Introduction

1.1 The Problem: Aligning Decision Procedures with Organizational Strategy

Every organization that makes collective decisions implicitly adopts a decision strategy — a set of values about what kinds of alternatives should win. A company that votes by plurality rule is implicitly optimizing for alternatives with the most enthusiastic core supporters, regardless of how many people find them unacceptable. A company that votes by approval rule is optimizing for alternatives that minimize rejection. Neither procedure makes this value choice explicit, and neither provides a principled way to navigate between the two extremes.

This paper addresses three tightly connected problems:

1. **Design problem:** how can an organization design a voting rule that provably implements a target welfare economics criterion — utilitarian, egalitarian, elitist, or Nash — while remaining computationally tractable at scale?
2. **Analysis problem:** given an existing voting rule, what welfare economics criterion does it implicitly implement? What types of alternatives does it systematically favor?
3. **Classification problem:** can we rigorously classify voting rules by their strategic positioning — the way they rank different types of alternatives — and use this classification to guide rule selection?

These three problems sit at the intersection of three research traditions: **Social Choice Theory (SCT)** studies the axiomatic properties of voting rules; **Welfare Economics (WE)** provides normative criteria for evaluating social outcomes in terms of utility, equity, and efficiency; and **Multi-Objective Optimization (MOO)** develops computational tools for navigating trade-offs between competing objectives. Each tradition addresses part of the problem; none addresses all three jointly.

This paper introduces the **Welfare Optimization (WO) framework**, which bridges all three traditions by embedding WE metrics directly into the parametrization of scalable voting rules. The framework's central object — the WO scoring rule — is a weighted mean of a vote profile vector that can be set to implement any of the four canonical WE evaluators by choosing appropriate weights. The strategic classification of any WO rule is then determined by how it orders nine archetypal alternatives, providing an ex ante decision tool for rule selection.

1.2 Key Definitions Upfront

To guide the reader, we define here the three central SCT concepts. Formal definitions appear in Sections 3 and 4.

- **Social Choice Theory (SCT):** the branch of economic theory that studies collective decision mechanisms aggregating individual preferences into a group decision. SCT characterizes rules by their axiomatic properties and identifies impossibility results (Arrow, 1951; Gibbard, 1973).

- **Welfare Economics (WE):** the branch of economics that evaluates social states based on individual utility or welfare. WE provides four canonical metrics: utilitarian sum, egalitarian leximin, elitist leximax, and Nash product.
- **Scalable Optimization:** in our context, a voting rule is scalable if its computation is linear in the number of voters and the ballot scale, and if it can be parallelized across independent sub-electorates with exact aggregation of partial results.

1.3 Why Connect These Three Traditions?

The connection between SCT, WE, and MOO is not arbitrary — it is forced by the structure of the problems organizations actually face. When a company must choose between competing projects, it simultaneously faces: **a social choice problem** (aggregating different experts' opinions into a group decision), **a welfare economics problem** (deciding whether to maximize total satisfaction, protect the least satisfied stakeholder, or identify the most enthusiastic supporters), and **an optimization problem** (finding the alternative that scores best under the chosen welfare criterion, across a potentially large set of alternatives).

SCT without WE provides axiomatic guarantees but no normative guidance on which type of alternative should win. WE without SCT provides normative criteria but lacks the axiomatic machinery to translate them into deployable procedures. MOO without SCT/WE provides optimization tools but no principled way to aggregate multi-stakeholder objectives. The WO framework closes all three gaps simultaneously.

2. Context and Related Work

2.1 The Correct Axiomatic Framework: Young (1975)

Arrow's impossibility theorem (1951) establishes that no **ordinal** preference aggregation rule satisfying five conditions (universality, Pareto, non-dictatorship, IIA, transitivity) can exist. This result applies to rules that take as input total orderings and produce a social ordering. The WO family takes as input **cardinal welfare evaluations** — scores in $\{1, \dots, R\} \cup \{\text{rejection}\}$ — and produces a real-valued score per alternative. This is a welfare aggregation problem, not a preference aggregation problem. Arrow's theorem does not apply.

The correct framework is **Young (1975)**, who characterizes **additive scoring rules** — rules of the form $\text{score}(a) = \sum_i s(r_i(a))$ for some score function s — by three axioms: anonymity, neutrality, and consistency. The WO family generalizes this class by allowing vector-valued score functions (distributions over ranks rather than point scores). Our Theorem 4.1 provides the exact characterization.

Remark — Terminological note: 'approval' vs. 'approbation'

Following the advice of referees with English-language expertise: this paper uses **approval** throughout (the term standard in the social choice literature, referring to the binary accepted/rejected distinction). The word *approbation*, though technically English, carries a stronger connotation and is not used in this domain. The **approval vector** (Definition 4.6) is named because it measures, for each rank threshold, the fraction of voters who approved the alternative — i.e., who did not reject it. See Section 4.3 for the formal reconciliation of this definition with the binary approval/rejection concept.

2.2 Related Work

- **Young (1975)**: characterizes additive scoring rules by anonymity, neutrality, consistency. Our WO family strictly generalizes his class to vector-valued score functions.
- **Brams & Sanver (2009)**: introduce Fallback Voting combining approval and ordinal preference — the direct precursor of the WO ballot. The WO ballot adds the gap symbol (?) and permits tied ranks (via $\sim$), as justified in Section 4.1.
- **Pivato (2013, 2014)**: develops welfare-based voting rules as statistical estimators. Closest theoretical relative in the literature.
- **Balinski & Laraki (2011)**: Majority Judgment based on median grades. Cross-paradigmatic: Balinski explicitly rejected welfarism as a voting model. WO-MJ comparisons are valid only with this caveat clearly stated (Remark 5.4).
- **d'Aspremont & Gevers (1977)**: formalize interpersonal utility comparability in welfare functions — reference for our Section 2.3 discussion.
- **Gibbard (1973) / Satterthwaite (1975)**: the strategy-proofness impossibility — directly addressed in Section 6.

2.3 Interpersonal Utility Comparability

Remark — Interpersonal Utility Comparability (IUC)

All utilitarian and Nash-welfare aggregations in this paper assume that voter utility values are interpersonally comparable — it is meaningful to sum or multiply them across voters. This assumption is standard in welfare economics applications (Sen, 1970; d'Aspremont & Gevers, 1977) and is justified in organizational settings by the shared ballot scale: since all voters use the same rank set $\{1, \dots, R\}$, a rank-1 vote from voter i carries the same formal utility weight R as a rank-1 vote from voter j .

Robustness if IUC is violated: the egalitarian (leximin) and elitist (leximax) comparators require only ordinal comparability within each voter's ballot, not across voters. If IUC is contested (e.g., in heterogeneous political electorates), these comparators remain valid. The utilitarian and Nash evaluators should not be used in such contexts.

2.4 Notation

Symbol	Type	Meaning
N	$\{1, \dots, n\}$	Voters (formerly: deciders / agents)
M	$\{a, b, \dots\}$	Alternatives (candidates)
T	$\{1, \dots, R\}$	Ranks (1 = highest expressed welfare, R = lowest among supported)
R	Integer ≥ 2	Ballot scale size
symbol(rejection)	Symbol	Explicit rejection of alternative by voter
P	$N \times M \rightarrow T \cup \{\text{rej.}\}$	Vote profile type
p	Instance of P	A specific vote profile; $p_i(a)$ = ballot of voter i for alternative a
$u_i(a)$	Real ≥ 0	Utility of voter i for alternative a
epsilon	Real	Neutral element: 0 (sum), 1 (product), +inf (min)
rate_p(a)	$[0, 1]^R$	Rate vector of alternative a in profile p
supp_p(a)	$[0, 1]^R$	Support vector (cumulative left-to-right = elitist measure)
appr_p(a)	$[0, 1]^R$	Approval vector (cumulative right-to-left = egalitarian measure)
omega	$T \rightarrow R$	Weighting function

3. Axiomatic Requirements

3.1 Scalability Criteria

Definition 3.1 — WO-IIA (Additive Separability)

A scoring rule ρ satisfies WO-IIA if and only if, for every pair of profiles p, p' and every alternative a :

$$(\text{for all } i \text{ in } N: p_i(a) = p'_i(a)) \text{ implies } \text{Eval}_{\rho}(p, a) = \text{Eval}_{\rho}(p', a)$$

Distinction from Arrow-IIA: Arrow-IIA requires that the social **ordering** of $\{a, b\}$ depends only on individual orderings of $\{a, b\}$. WO-IIA requires that the **score** of a depends only on ballots for a . These are distinct: WO-IIA is satisfied trivially by any additive scoring rule; Arrow-IIA is not satisfiable by any non-dictatorial ordinal rule with transitive output (Arrow, 1951). Presenting WO-IIA as 'satisfying Arrow's IIA' would be misleading; we do not make this claim.

Definition 3.2 — Independence of Unconcerned Voters (IUV)

ρ satisfies IUV if and only if, for any profile p and modification p' where $p'_i(a) = p_i(a)$ for all a and some subset S of voters, and $p'_i(a) = p_i(a)$ for all i not in S :

for all a in M : $Eval_rho(p', a) = Eval_rho(p, a)$ for all a whose ballots from S are unchanged

Definition 3.3 — Independence of Common Scale (ICS)

A scoring rule ρ satisfies ICS if and only if, for any profile p and any positive scalar $\lambda > 0$, the rescaled profile p^λ defined by $u^\lambda_i(a) = \lambda * u_i(a)$ satisfies:

for all a, b in M : $Eval_rho(p^\lambda, a) \geq Eval_rho(p^\lambda, b)$ iff $Eval_rho(p, a) \geq Eval_rho(p, b)$

Critical assumption: rescaling utilities by λ does not change the rank assignments. Formally: if $u_i(a) = R - p_i(a) + 1$ and $u^\lambda_i(a) = \lambda * (R - p_i(a) + 1)$, then $p_i(a)$ is determined by the **ordinal** position, not the cardinal utility value. The rank $p_i(a) = r$ means voter i places a at position r in their ranking; multiplying by λ scales the utility representation but not the position. Lemma 3.1 formalizes this.

Proposition 3.1 — Rank invariance under utility rescaling

For any voter i , alternative a , and positive scalar $\lambda > 0$: if $u_i(a) = R - p_i(a) + 1$ and $u_i(b) = R - p_i(b) + 1$, then $u^\lambda_i(a) > u^\lambda_i(b)$ if and only if $u_i(a) > u_i(b)$, if and only if $p_i(a) < p_i(b)$.

Proof. $\lambda > 0$ is a positive scalar, so it preserves all strict inequalities: $\lambda * u > \lambda * v$ iff $u > v$. Since $p_i(a) = R - u_i(a) + 1$, the rank ordering is the unique monotone transformation of utility, and is preserved by any positive rescaling. Therefore $rate_p^\lambda(a)[r] = rate_p(a)[r]$ for all r . Since rate vectors are unchanged, all downstream vectors (cumulative, support, approval) are unchanged, and all WO scores are unchanged. ICS holds. (square)

Definition 3.4 — Summability

ρ is summable if and only if for any profile partition $N = N_1 \cup N_2$:

for all a in M : $Eval_rho(p, a) = Eval_rho(p_1, a) + Eval_rho(p_2, a)$

Threshold filters break Summability: a high-pass filter HP_θ is non-linear at the threshold boundary. Counter-example: if $rate_{\{N_1\}}(a)[r] = 0.3 < \theta = 0.4$ and $rate_{\{N_2\}}(a)[r] = 0.3 < \theta$, but $rate_N(a)[r] = 0.3 \geq \theta$ (when mixing), the filtered values differ. Summability requires linearity; threshold filters violate it. All other criteria remain valid with threshold filters.

Definition 3.5 — Resolvability

Form 1 (constructive): for every tied profile, there exists an additional ballot that breaks the tie uniquely.

Form 2 (asymptotic): under any non-atomic probability distribution D on individual ballots, the tie probability satisfies:

$Pr_D[\text{exists } a, b \text{ in } M : Eval_rho(p, a) = Eval_rho(p, b)] \rightarrow 0 \text{ as } n \rightarrow \infty$

Proof of Form 2. Under any non-atomic distribution D , the probability that voter i casts any specific ballot b is 0. The score $\text{Eval_rho}(p, a)$ is a sum of n independent bounded random variables. By the law of large numbers, this sum concentrates around its expectation at rate $1/\sqrt{n}$. Two such sums (for alternatives a and b) are equal with probability tending to 0 under any non-atomic D , since this event has measure 0 in the product space of n voter ballot draws. (square)

3.2 Resilience Criteria

Definition 3.6 — Anonymity

Rho is anonymous iff for every permutation π of voters: $\text{Eval_rho}(p, a) = \text{Eval_rho}(p \circ \pi, a)$ for all a .

Definition 3.7 — Neutrality

Rho is neutral iff for every permutation γ of alternatives, the transformed profile γ^*p satisfies: $\text{Eval_rho}(\gamma^*p, \gamma(a)) = \text{Eval_rho}(p, a)$ for all a .

Definition 3.8 — Consistency

Rho is consistent iff for any two disjoint electorates N_1, N_2 with profiles p_1, p_2 over the same alternatives: if $\text{Eval_rho}(p_1, a) > \text{Eval_rho}(p_1, b)$ and $\text{Eval_rho}(p_2, a) \geq \text{Eval_rho}(p_2, b)$, then $\text{Eval_rho}(p_1 \cup p_2, a) > \text{Eval_rho}(p_1 \cup p_2, b)$. This is the critical axiom in Young's (1975) characterization: together with anonymity and neutrality, it completely characterizes additive scoring rules.

Definition 3.9 — Monotonicity

Rho is monotone iff for any improvement of alternative a 's ballot by voter i (from rank r to $r' < r$, or from rejection to any rank r), the new profile p' satisfies: $\text{Eval_rho}(p', a) \geq \text{Eval_rho}(p, a)$ and $\text{Eval_rho}(p', b) = \text{Eval_rho}(p, b)$ for all $b \neq a$.

4. The WO Rule Family

4.1 The WO Ballot — With Justification of the Gap Symbol

Definition 4.1 — Welfare Optimization (WO) Ballot

Let $M = \{a_1, \dots, a_m\}$ and $T = \{1, \dots, R\}$. A WO ballot is a function $b: M \rightarrow T \cup \{\text{rejection}\}$. The utilities are $u_i(a) = R - p_i(a) + 1$ if $p_i(a) \in T$, and $u_i(a) = \epsilon$ (neutral element) if $p_i(a) = \text{rejection}$.

The graphical representation uses the symbols: $>$ (rank change), $\sim$ (equal rank), $|$ (support/rejection separator), $,$ (between rejected alternatives). The symbol $?$ allows a voter to skip a rank level without naming an alternative at that level.

Justification of the gap symbol $?$: the symbol $?$ addresses a genuine expressiveness need. Consider a voter who strongly supports a (rank 1) and weakly supports b (rank 3), with no alternative at rank 2 in their mind. Forcing them to choose between ' a at rank 1, b at rank 2' (misrepresenting b as their second-best thought) or ' a at rank 1, b at rank 1' (misrepresenting equal enthusiasm) distorts the ballot. The $?$ symbol allows the expression ' $a > ? > b$ ', meaning a is first, b is third, with rank 2 intentionally empty. This adds expressiveness without adding cardinality: the gap is ordinal (a rank position is skipped), not cardinal (no numerical value is attached). The voter still ranks b at an integer rank 3; the $?$ is simply a positional placeholder.

Relationship to Brams-Sanver (2009): the Brams-Sanver Fallback Voting ballot allows voters to rank disapproved alternatives. The WO ballot does not require this (disapproved alternatives go to the right of $|$, unranked). The $?$ symbol and the $\sim$ symbol are WO-specific additions that increase expressiveness. The rest of the paper excludes $?$ for formal results, but the symbol remains valid in practice.

4.2 Vote Profile and Alternative Profiles

Definition 4.2 — Vote Profile

A vote profile is a function $p: N \times M \rightarrow T \cup \{\text{rejection}\}$, assigning to each voter i in N and alternative a in M a ballot value $p_i(a)$. The approvers of a in p are: $\text{approvers}_p(a) = \{i \in N : p_i(a) \neq \text{rejection}\}$.

Definition 4.3 — Rate Vector

$$\text{rate}_p(a)[r] = |\{i \in \text{approvers}_p(a) : p_i(a) = r\}| / |N|$$

Voters who rejected a are excluded from the numerator. The sum $\sum_{r \in T} \text{rate}_p(a)[r] = |\text{approvers}_p(a)| / |N| \leq 1$; the gap is the rejection rate.

Definition 4.4 — Support Vector

$\text{cumul}_p(a)[r] = \sum_{k=1}^r \text{rate}_p(a)[k]$. The support vector: $\text{supp}_p(a)[r] = \text{cumul}_p(a)[r]$.

$\text{supp}_p(a)[r]$ is the fraction of all voters who placed a among their top r choices. High values at low r indicate intense enthusiasm — the elitist signal.

Definition 4.5 — Approval Vector

$\text{appr}_p(a)[r] = \text{cumul}_p(a)[R - r + 1]$.

$\text{appr}_p(a)[r]$ is the fraction of all voters who did not place a in the bottom r positions (i.e., who approved a or ranked it better than the bottom r positions). High values at low r indicate broad acceptance — the egalitarian signal.

4.3 Reconciling the Approval Vector with the Binary Approval Concept

Response to reviewer — Approval vector paradox — Response to ARTINT Reviewer 1

The apparent paradox. Reviewer 1 observed: 'The approval vector is well-defined but it is contradicting to the notion of approval previously referred in the text. The authors state that approval is related to whether an alternative is approved or rejected (which they highlight as one of the advantages of their proposed ballot) but in the definition they use the rank to define the approval vector.'

Resolution. The apparent contradiction dissolves once we observe that the approval vector is not the **binary** approval concept — it is a **graded extension** of it. Specifically:

Binary approval (as in approval voting): voter i approves a iff $p_i(a) \neq \text{rejection}$. This is captured by $\text{appr}_p(a)[R] = |\text{approvers}_p(a)| / |N|$ — the last component of the approval vector, which counts the fraction of non-rejectors regardless of rank.

Graded approval: $\text{appr}_p(a)[r]$ counts, for each rank threshold r , the fraction of voters who did not place a in the r lowest supported positions. As r decreases from R to 1, the criterion tightens: $\text{appr}_p(a)[1]$ is the fraction of voters who placed a at the very best rank (pure elitist information), while $\text{appr}_p(a)[R]$ is the fraction who simply approved a (pure binary approval information).

The approval vector therefore **generalizes** binary approval: the WO rule with lexicographic weights on the approval vector exactly reproduces approval voting (Theorem 5.5), while finer weight functions capture the grade distribution within approved alternatives. This is a feature, not a contradiction: the ballot captures both binary and graded approval information, and the weight function selects how much weight to give each.

Definition 4.6 — Threshold Filters

High-pass filter HP_{θ} : $\text{HP}_{\theta}(v)[r] = v[r]$ if $v[r] \geq \theta$, else 0.

Low-pass filter LP_{θ} : $\text{LP}_{\theta}(v)[r] = \min(v[r], \theta)$ applied entry-wise.

Formal properties: both HP_{θ} and LP_{θ} are functions $[0,1]^R \rightarrow [0,1]^R$. They are monotone non-decreasing component-wise (hence preserve Monotonicity with positive weights) but non-linear (hence break Summability).

Definition 4.7 — Consensuality Vector

Let $(+)$ be a binary operator on $[0,1]$ that is commutative, associative, non-decreasing in both arguments, and satisfies $(+)(0,0)=0$ and $(+)(1,1)=1$. Let $\tau = (\tau_s, \tau_a)$ specify optional filters. The consensuality vector is:

$$\text{cons}_p^{\tau}(a)[r] = \text{supp}_p^{\tau_s}(a)[r] (+) \text{appr}_p^{\tau_a}(a)[r]$$

Valid operators: arithmetic mean $(x+y)/2$, geometric mean $\sqrt{xy}$, $\min(x,y)$, product xy .

4.4 The WO Scoring Rule and Representation Theorem

Definition 4.8 — WO Scoring Rule and the Score Function Space

A WO scoring rule is a pair (π, ω) where $\pi_p(a)$ in $[0,1]^R$ is one of $\{\text{rate}_p(a), \text{supp}_p(a), \text{appr}_p(a), \text{cons}_p^{\tau}(a)\}$ and $\omega: T \rightarrow R$ is the weighting function. The score is:

$$\text{Eval}_{\{(\pi, \omega), p \}}(a) = \sum_{r=1}^R \omega(r) * \pi_p(a)[r]$$

The score function space S_{WO} . The score function underlying any WO rule is a map $s: T \cup \{\text{rejection}\} \rightarrow R$ lying in the space:

$$S_{\text{WO}} = \{ s: T \cup \{\text{rejection}\} \rightarrow R : s(1) \geq s(2) \geq \dots \geq s(R) \geq 0 \text{ and } s(\text{rejection}) = 0 \}$$

The three conditions are: (C1) **welfare-monotonicity** — $s(r) \geq s(r')$ for all $r < r'$ (better rank $\Rightarrow$ higher score); (C2) **non-negativity of supported alternatives** — $s(R) \geq 0$ (worst supported rank is at least as good as rejection); (C3) **rejection neutrality** — $s(\text{rejection}) = \epsilon = 0$ (rejection contributes nothing to the additive aggregate). These three conditions are necessary and sufficient for the WO representation, as established in Theorem 4.1.

Constraints on ω : $\omega(r) = |N| * s(r)$, hence: $\omega(r) \geq \omega(r')$ for $r < r'$ (weakly decreasing in rank = weakly increasing in welfare), and $\omega(r) \geq 0$ for all r . When π is the rate vector, these constraints imply ω weakly increasing; when π is the cumulative support or approval vector, the structural monotonicity is already built into the vector, and $\omega > 0$ suffices.

Theorem 4.1 — WO Representation Theorem — Complete Proof with Functional Characterization

Statement. The following are equivalent for a scoring rule F :

(i) F is a WO scoring rule (Definition 4.8) — i.e., $\text{Eval}_F(p,a) = \sum_r \omega(r) * \pi_p(a)[r]$ for some π in $\{\text{rate}, \text{supp}, \text{appr}, \text{cons}\}$ and $\omega: T \rightarrow R$.

(ii) F satisfies anonymity, neutrality, consistency, WO-IIA, and monotonicity (Definitions 3.1, 3.6–3.9), **and** its underlying score function lies in S_WO (Definition 4.8): s in S_WO , meaning $s(1) \geq s(2) \geq \dots \geq s(R) \geq s(\text{rejection}) = 0$.

Necessity of each condition in S_WO :

C1 (welfare-monotonicity): if $s(r) < s(r')$ for some $r < r'$, then a voter improving from rank r' to rank r would **decrease** F 's score, violating Monotonicity (Def. 3.9). $C1$ is necessary.

C2 (non-negativity): if $s(R) < 0$, then approving an alternative at the worst rank would decrease its score compared to rejection, creating a negative marginal contribution. This contradicts Monotonicity (approving at any rank should be at least as good as rejecting). $C2$ is necessary.

C3 (rejection neutrality): if $s(\text{rejection}) \neq 0$, then rejection contributes a non-zero amount to the aggregate, making the rule non-separable in the WO-IIA sense (the rejection count would affect scores). $C3$ is necessary.

Proof (i implies ii). Given a WO rule (π, ω) : anonymity holds because $\text{rate}_\pi(a)[r]$ is symmetric in voter identities. Neutrality holds because the score function operates on rank positions, label-free. Consistency holds by the convex-combination structure of merged rate vectors. WO-IIA holds by construction (each alternative's vector uses only its own ballots). Monotonicity holds with ω non-decreasing (rate vector) or ω positive (cumulative vectors). The score function $s(r) = \omega(r)/|N|$ lies in S_WO by the constraints on ω . (square)

Proof (ii implies i). Given F in condition (ii): by Young (1975), anonymity + neutrality + consistency implies $\text{Eval}_F(p, a) = \sum_i s(\pi_i(a))$ for some $s: T \cup \{\text{rejection}\} \rightarrow \mathbb{R}$. Grouping by rank:

$$\text{Eval}_F(p, a) = |N| * \sum_{\{r \text{ in } T\}} s(r) * \text{rate}_\pi(a)[r] + |N| * s(\text{rejection}) * \text{rejection_rate}_\pi(a)$$

By $C3$, $s(\text{rejection}) = 0$, so the rejection term vanishes. Setting $\omega(r) = |N| * s(r)$ gives the WO rate-vector form. By $C1$, $s(r) \geq s(r')$ for $r < r'$, so ω is weakly decreasing in rank (weakly increasing in welfare). By $C2$, $\omega(r) \geq 0$. This is exactly the WO constraint set. Therefore $F = \text{WO}(\text{rate}, \omega)$. (square)

Corollary (functional space). The WO family is the exact preimage of S_WO under Young's (1975) score-function map: $\text{WO} = \{ F : s_F \text{ in } S_WO \}$. The support vector representation (ω not necessarily weakly increasing) and approval vector representation are equivalent reformulations within S_WO , not separate families.

Example 4.1 — Representation theorem verification

Profile: $N=\{1,2,3\}$, $M=\{a,b,c\}$, $T=\{1,2,3\}$, arithmetic weights $\omega(r)=1$, π = support vector.

Agent 1: $a=r1$, $b=r2$, $c=\text{rejection}$. Agent 2: $a=r2$, $b=r1$, $c=r3$. Agent 3: $a=\text{rejection}$, $b=r1$, $c=r2$.

$$\text{rate}_\pi(a)=[1/3, 1/3, 0], \text{ supp}_\pi(a)=[1/3, 2/3, 2/3], \text{ Eval}(a)=1/3+2/3+2/3=5/3$$

$$\text{rate}_p(b)=[2/3, 1/3, 0], \text{supp}_p(b)=[2/3, 1, 1], \text{Eval}(b)=2/3+1+1=8/3$$

$$\text{rate}_p(c)=[0, 1/3, 1/3], \text{supp}_p(c)=[0, 1/3, 2/3], \text{Eval}(c)=0+1/3+2/3=1$$

Score function: $s(r) = R-r+1 = 3-r+1$. $s(1)=3$, $s(2)=2$, $s(3)=1$, $s(\text{rej})=0$. Young's form: $\text{Eval}_F(p,b) = (1/3)*[3*2 + 2*1 + 1*0 + 0*0] * 3 = 8/3$. Matches. (✓)

5. Complete Axiomatic Proofs

5.1 Proofs of Anonymity, Neutrality, Consistency, Monotonicity

Theorem 5.1 — All WO rules satisfy anonymity, neutrality, consistency, and monotonicity

Anonymity: $\text{rate}_p(a)[r] = |\{i : p_i(a)=r\}|/|N|$ is symmetric in voter identities. Permuting voters gives the identical count. (square)

Neutrality: the score function operates on rank values (integers 1 to R), not on alternative labels. Relabeling alternatives by any permutation γ produces the same distribution of rank values, hence the same rate vector and score. (square)

Consistency: as shown in the proof of Theorem 4.1 (implication i=>ii), consistency follows from the convex-combination structure of the merged rate vector. (square)

Monotonicity (rate vector, weights non-decreasing): improving voter i from rank r' to $r < r'$ shifts $\text{rate}_p(a)[r']$ by $-1/|N|$ and $\text{rate}_p(a)[r]$ by $+1/|N|$. The change in score is $(\omega(r) - \omega(r'))/|N| \geq 0$ since $r < r'$ and ω is non-decreasing. (square)

Monotonicity (support/approval/consensuality vectors, weights positive): any ballot improvement produces a component-wise dominant cumulative vector (the approval rate at every threshold can only increase or stay). With positive weights, the weighted sum can only increase. This applies to high-pass and low-pass threshold filters: HP_θ and LP_θ are both monotone non-decreasing functions, so the composition with a positive-weighted sum remains monotone. (square)

5.2 ICS — With Explicit Rank-Invariance Lemma

Theorem 5.2 — All WO rules satisfy WO-IIA, ICS, IUUV, and conditional Summability

WO-IIA: each alternative's rate vector is computed solely from ballots for that alternative. Modifying any ballot for $b \neq a$ has zero effect on $\text{rate}_p(a)$, hence zero effect on $\text{Eval}(a)$. (square)

ICS: by Proposition 3.1, rescaling utilities by any $\lambda > 0$ leaves all rank assignments unchanged, hence all rate vectors unchanged, hence all WO scores unchanged (up to the same positive multiplicative constant λ , which preserves all score rankings). (square)

IUV: a direct consequence of WO-IIA applied to the unchanged sub-group of voters. (square)

Summability (without threshold filters): $\text{rate}_{\{N\}}(a)[r] = (n_1/n) \cdot \text{rate}_{\{N_1\}}(a)[r] + (n_2/n) \cdot \text{rate}_{\{N_2\}}(a)[r]$. The weighted sum is linear in this expression, establishing Summability up to the normalization factor n (irrelevant for ranking). (square)

5.3 Resolvability — Formal Proof

Theorem 5.3 — Resolvability — both forms

Form 1 (constructive): given a tied profile p where $\text{Eval}(p,a) = \text{Eval}(p,b)$, add a single ballot that ranks a strictly higher than b (or approves a and rejects b). This shifts $\text{rate}_p(a)$ and $\text{rate}_p(b)$ in favor of a by exactly $1/|N+1|$ in the relevant position. For non-lexicographic weights, this generically breaks the tie. For lexicographic weights, the argument requires adding a ballot at the first discriminating rank. (square)

Form 2 (asymptotic): let D be any non-atomic probability distribution on the ballot space T^M union $\{\text{rejection}\}^M$. Under D , the probability that any specific rank assignment occurs is 0. The score $\text{Eval}_{\rho}(p, a) = (1/|N|) \cdot \sum_i f(p_i(a))$ where f is a bounded function. By the central limit theorem, $(1/n) \cdot \sum_i f(p_i(a))$ converges almost surely to $E_D[f(p_i(a))]$. For two distinct alternatives a, b with distinct expected scores $E_D[f(p_i(a))] \neq E_D[f(p_i(b))]$, the event $\{\text{Eval}(p,a) = \text{Eval}(p,b)\}$ has probability converging to 0 at rate $1/\sqrt{n}$. If $E_D[f(p_i(a))] = E_D[f(p_i(b))]$ (same expected scores), ties are resolved by the Resolvability Form 1 criterion. (square)

5.4 The Four WE Metrics — Complete Proofs

Theorem 5.4 — Utilitarian = arithmetic weights on support or approval

For all profiles p and alternatives a : $\text{Eval}_{\{(\text{supp}, \omega_{\text{arith}})\}}(p, a) = \text{util}_p(a) / |N|$, where $\text{util}_p(a) = \sum_{i \in N} u_i(a)$.

Proof. $\sum_r \text{supp}_p(a)[r] = \sum_r \sum_{k \leq r} \text{rate}(a)[k] = \sum_k (R-k+1) \cdot \text{rate}(a)[k] = \text{util}_p(a) / |N|$. The double sum reversal: each term $\text{rate}(a)[k]$ appears for all $r \geq k$, i.e., $(R-k+1)$ times. Since $u_i(a) = R-p_i(a)+1$, the result follows. (square)

Theorem 5.5 — Egalitarian leximin is equivalent to lexicographic comparison of the approval vector

For all profiles p and alternatives a, b : $a \succ_{\text{leximin}} b$ if and only if $\text{appr}_p(a) \succ_{\text{lex}} \text{appr}_p(b)$.

Proof. Leximin sorts utilities from lowest to highest and compares index by index. The minimum utility is 0 for rejected alternatives. So $\text{appr}_p(a)[R] = |\text{approvers}_p(a)|/|N|$ = fraction of non-rejectors. Higher $\text{appr}_p(a)[R]$ means fewer rejectors means higher leximin first entry. Inductively, $\text{appr}_p(a)[r]$ for decreasing r corresponds to the increasing-utility sort of the utility vector. The equivalence follows from the bijection between the approval vector and the sorted utility vector. (square)

Theorem 5.6 — Elitist leximax is equivalent to lexicographic comparison of the support vector

For all profiles p and alternatives a, b : $a \succ_{\text{leximax}} b$ if and only if $\text{supp}_p(a) \succ_{\text{lex}} \text{supp}_p(b)$.

Proof. Leximax sorts from highest utility to lowest. The maximum utility R corresponds to rank 1. $\text{supp}_p(a)[1] = \text{rate}_p(a)[1]$ = fraction of voters placing a at rank 1 = fraction at utility R . Higher $\text{supp}_p(a)[1]$ means more top-rank voters means higher leximax first entry. Inductively, $\text{supp}_p(a)[r]$ corresponds to the decreasing-utility sort. Note: this proof is entirely independent of Theorem 5.5; no cross-reference is needed. (square)

Theorem 5.7 — Nash welfare = geometric weights on rate vector

For all profiles p and alternatives a : $\log(\text{nash}_p(a)) = \sum_r \log(R-r+1) * \text{rate}_p(a)[r]$, where $\text{nash}_p(a) = \text{product}_{\{i: p_i(a) \neq \text{rejection}\}} u_i(a)$.

Proof. Grouping by rank: $\text{nash}_p(a)$ proportional to $\text{product}_{\{r \in T\}} (R-r+1)^{|N| * \text{rate}_p(a)[r]}$. Taking logarithm (strictly increasing, preserving order): $\log(\text{nash}_p(a)) = \sum_r |N| * \text{rate}_p(a)[r] * \log(R-r+1)$. Up to the constant $|N|$ (irrelevant for ranking by ICS): $\text{Eval}_{\{(\text{rate}, \text{omega_geom})\}}(p, a)$. (square)

Remark — Majority Judgment — cross-paradigmatic comparison

Balinski & Laraki (2011) explicitly rejected welfarism as a model for voting. Within the WO framework, MJ is modeled as lexicographic weights on the 50%-filtered support vector. This model captures the median-seeking behavior of MJ but not its absolute-grade semantics. The strategic comparison (MJ is weakly approving within the WO taxonomy) is valid as a model prediction but should not be interpreted as a claim about MJ's original design intent.

5.5 Complexity

Theorem 5.8 — $O(n + R)$ complexity, with $O(\sqrt{n * R})$ distributed extension

For a single alternative, computing any WO score requires $O(n + R)$ time and $O(R)$ space. Distributed over k parallel nodes with Summability: $O((n/k) + k * R)$ time per alternative. Optimal at $k = \sqrt{n/R}$ nodes, giving $O(\sqrt{n * R})$ wall-clock time per alternative.

Proof. Step 1: count ballots into rate vector — $O(n)$. Step 2: cumulative prefix sum — $O(R)$. Step 3: threshold filter (if any) — $O(R)$. Step 4: weighted sum — $O(R)$. Total: $O(n + R)$. Distribution: each node processes n/k ballots in $O(n/k)$; merge requires $O(k \cdot R)$ weighted sum combination. Minimizing $O(n/k + k \cdot R)$ over k : derivative = $-n/k^2 + R = 0 \Rightarrow k = \sqrt{n/R}$. (square)

6. Strategic Manipulation — Complete Analysis

6.1 Impossibility Boundary

Gibbard-Satterthwaite Theorem (reference result)

Any deterministic social choice function $f: P \rightarrow M$ with $|M| \geq 3$ that is onto (every alternative can be elected) and non-dictatorial is **manipulable**: there exist voter i , profile p , and misreport p'_i such that $f(p'_i, p_{-i})$ is strictly preferred by voter i under their true preferences. WO rules are manipulable for $|M| \geq 3$. This is an inherent property of all non-trivial deterministic voting rules, not a specific deficiency of WO.

6.2 Dominant Strategies — Per Rule Class

Theorem 6.1 — Dominant strategies under WO rules

Under strongly supportive rules (leximax/plurality-type): the dominant strategy for voter i who prefers alternative a is to place a at rank 1, regardless of true enthusiasm, and reject all alternatives they do not prefer. This replicates the strategic incentive of plurality rule: **bullet voting** — concentrate all support on one candidate.

Under strongly approving rules (leximin/approval-type): the dominant strategy is **sincere approval** — approve all alternatives preferred over the expected winner, reject all others. Since the rule is blind to relative ranking among approved alternatives, there is no incentive to misrank within the approved set. This property is approval voting's well-known strategic simplicity.

Under utilitarian rules (Borda/arithmetic): the dominant strategy may involve **rank compression** — inflating the preferred alternative and deflating the expected winner. However, the marginal gain from any single rank change is bounded by $\omega(r) - \omega(r+1)$. For arithmetic weights, this is 0 — confirming that individual rank changes do not affect Borda score, only the distribution of approvals/rejections matters. Borda is vulnerable to strategic rejection (rejecting strong competitors) rather than rank manipulation.

6.3 Nash Equilibrium Analysis

We characterize the Nash equilibria of the voting game induced by each WO rule class, where each voter's strategy is their ballot and their payoff is 1 if their preferred alternative wins, 0 otherwise.

Proposition 6.1 — Nash equilibria under plurality-type WO rules

Under any strongly supportive WO rule, a strategy profile p^* is a Nash equilibrium if and only if: (i) all voters who prefer the equilibrium winner a^* place a^* at rank 1 (no voter has incentive to place a^* lower), and (ii) no voter can profitably deviate to make a different alternative win. Formally: p^* is a Nash equilibrium iff it is a **single-peaked profile** where all supporters of the winning alternative are at rank 1, and no coalition of less than $n/2$ voters can shift the outcome.

Proof sketch. Any voter placed at rank 1 already maximizes their contribution to the winning alternative's score. Deviating to a lower rank strictly reduces this contribution, making the deviation unprofitable if a^* is currently winning. The equilibrium condition reduces to: $|\{i: p^*_i(a^*) = 1\}| > |\{i: p^*_i(b) = 1\}|$ for all $b \neq a^*$, which is the standard Duverger equilibrium condition under plurality. (square)

Proposition 6.2 — Nash equilibria under approval-type WO rules

Under any strongly approving WO rule, **sincere approval** is a weakly dominant strategy — it weakly dominates all alternative strategies regardless of other voters' strategies. Therefore, the profile where every voter approves their genuinely preferred alternatives and rejects all others is always a Nash equilibrium.

This is the key strategic advantage of approval-type rules over plurality-type rules: sincere voting is an equilibrium, eliminating the coordination problem that plagues plurality elections (Duverger's law effect).

Proposition 6.3 — Nash equilibria under utilitarian (Borda-type) WO rules

Under any weakly consensual WO rule (arithmetic weights on support or approval), the Nash equilibrium set is larger and more complex than under pure supportive or approving rules. The following characterization holds:

(a) Sincere voting is not generally a Nash equilibrium. A voter whose true preferences are $a > b > c$ may profitably deviate by rejecting b (the expected runner-up), converting their ballot from ' $a > b > c$ ' to ' $a > c \mid b$ '. This **rank compression** strategy shifts weight from b 's score to c 's score, potentially allowing a to win by pulling votes from b rather than building its own support.

(b) The manipulation threshold. The gain from rank compression is bounded: for arithmetic weights, the benefit of demoting b from rank r to rejection is exactly $\omega(r) \cdot \text{rate_increase_penalty} / |N|$, where $\text{rate_increase_penalty} = 1$ (one voter changes ballot). The deviation is profitable only when the margin between a and b is smaller than this bound. Since $\omega(r) = 1$ for all r under arithmetic weights, the manipulation gain per voter is at most $1/|N|$ — very small for large electorates.

(c) Approximate equilibria. As $n \rightarrow \infty$, the gain from any single-voter deviation under Borda-type rules tends to 0 (by the $1/|N|$ bound). Therefore, for large electorates, sincere voting is an ***epsilon-Nash equilibrium*** for arbitrarily small $\epsilon > 0$. This provides a formal sense in which Borda-type rules are 'approximately strategy-proof' in large elections.

Proposition 6.4 — Nash equilibria under consensual (Nash-welfare/geometric) WO rules

Under Nash-welfare rules (geometric weights on rate vector), strategic incentives are the most complex due to the multiplicative structure. The key result is:

Theorem (Manipulation resistance of Nash welfare). For any voter i with utility $u_i(a)$ for the preferred alternative, the gain from any manipulation (changing rank or approval of any alternative $b \neq a$) is bounded above by:

$$gain_i \leq \log(1 + 1/n) / \log(nash_p(a))$$

Since $\log(1 + 1/n) \rightarrow 0$ as $n \rightarrow \infty$, and $\log(nash_p(a))$ is bounded away from 0 for any non-degenerate profile, the manipulation gain under Nash welfare rules **converges to 0** faster than under Borda-type rules. This confirms the simulation result (Section 6.4) that Nash-welfare rules have the lowest manipulation frequency.

Intuition. The geometric weighting amplifies the cost of any deviation that reduces the product of utilities. A voter who misreports to help a already at a high utility level gains very little (diminishing returns in the log-product), while a voter who misreports to penalize a strong competitor may inadvertently damage the Nash product in unpredictable ways.

6.4 Manipulation Frequency — Simulation with Two Preference Distributions

To assess robustness to the preference distribution, we run the manipulation simulation under two contrasting distributions: (U) **Uniform** — each voter assigns a random permutation of ranks uniformly at random; and (B) **Bimodal/polarized** — voters are split into two camps of size $n/2$ each, with opposite strong preferences (camp 1 places A,B in top 2 ranks and rejects C,D,E; camp 2 does the reverse). $n=21$, $m=5$, $R=5$, 10,000 profiles per cell:

Rule	U: manip. %	U: outcome %	B: manip. %	B: outcome %	Key observation
Plurality	38.2%	21.4%	52.1%	31.8%	<i>Polarization sharply increases strategic voting</i>
Approval	14.7%	8.3%	9.4%	4.1%	<i>Manipulation DECREASES under polarization</i>
Borda	29.1%	14.8%	38.7%	22.3%	<i>Moderate increase; rank compression more lucrative</i>

Nash / geometric	11.3%	6.1%	8.9%	4.7%	<i>Most robust: lower under both distributions</i>
Geom. Consensual	17.4%	9.2%	14.1%	7.6%	<i>Intermediate robustness</i>

Table: Manipulation simulation under uniform (U) and bimodal/polarized (B) distributions. $n=21$, $m=5$, $R=5$, 10,000 profiles each. Seed=42.

Key finding. The bimodal distribution reveals a striking divergence: approval-type rules become *less* vulnerable to manipulation under polarization (9.4% vs 14.7%), while plurality-type rules become *more* vulnerable (52.1% vs 38.2%). This confirms the strategic advantage of approval rules in politically polarized contexts — the regime where they are most commonly deployed. Nash-welfare rules maintain the lowest manipulation rates under both distributions, consistent with Proposition 6.4.

7. Archetype Taxonomy

7.1 The Nine Archetypes

The nine archetypes are defined by crossing three tiers of support with three tiers of approval (Table 1). Tier boundaries: Low = $[0, 0.35)$, Middle = $[0.35, 0.65)$, High = $[0.65, 1]$.

	Low Support	Middle Support	High Support
High Approval	Mitigated	Viable	Consensual
Middle Approval	Unsatisfying	Compromise	Performant
Low Approval	Dominated	Unpopular	Cleaving

7.2 The Mitigated Archetype Paradox — Resolution

Response to reviewer — Mitigated archetype paradox — Response to ARTINT Reviewer 1

The apparent paradox. Reviewer 1 observed: 'The Mitigated type refers to alternatives with high approval and low support. When an alternative has low support, it is ranked last by many voters. Rational thought implies that an alternative ranked last is more likely to also be rejected.'

Resolution. The paradox arises from conflating *low support* (few voters place the alternative at the top ranks) with *being ranked last*. These are different. Low support means: few voters place the alternative among their top choices — but many may place it in the middle ranks, which is compatible with approving it (not rejecting it).

Formally: an alternative has high approval ($\text{appr}_p(a)[R]$ close to 1) and low support ($\text{supp}_p(a)[1]$ close to 0) when: (i) very few voters place a at rank 1 (low enthusiasm), but (ii) very few voters also reject a . This is the 'acceptable-but-not-exciting' profile: a is everyone's third or fourth choice, but no one strongly opposes it. This is **not** paradoxical — it is the standard profile of a consensus compromise candidate.

Example. $R=5$ ranks, 10 voters. All 10 voters give a a rank 3. Then: $\text{supp}_p(a)[1]=0$ (no one places a first); $\text{appr}_p(a)[R=5]=1$ (all voters approved a). This is precisely the Mitigated archetype. No voter is ranking a last; they are all ranking it in the middle.

7.3 Fuzzy Protocol — Complete Specification

Definition 7.1 — Fuzzy Tier Membership Functions

For x in $[0,1]$: $\mu_{\text{Low}}(x) = 1$ for $x \leq 0.20$; linearly decreasing to 0 at $x=0.35$; 0 for $x > 0.35$. $\mu_{\text{Middle}}(x) = 0$ for $x \leq 0.30$; linearly increasing to 1 at $x=0.40$; 1 for x in $[0.40,0.60]$; linearly decreasing to 0 at $x=0.70$; 0 for $x > 0.70$. $\mu_{\text{High}}(x) = 0$ for $x < 0.60$; linearly increasing to 1 at $x=0.75$; 1 for $x \geq 0.75$. Archetype membership: $\mu(s, a, \text{tier}_S, \text{tier}_A) = \min(\mu_{\{\text{tier}_S\}}(s), \mu_{\{\text{tier}_A\}}(a))$.

These are standard triangular/trapezoidal membership functions used in fuzzy logic (Zadeh, 1997).

Definition 7.2 — Archetype Profile Construction — Full Specification

Parameters: $n_{\text{AS}} = 10^7$ alternative samples; $n_{\text{voters}} = 10^4$ per sample; $R = 90$ ranks; $\sigma = 0.20$ (Gaussian std. dev. in $[0,1]$ scale); random seed = 42.

Step 1 — Sample: for $k = 1, \dots, n_{\text{AS}}$: draw $(\mu_S^k, \mu_A^k) \sim \text{Uniform}([0,1]^2)$. Set Gaussians $S^k \sim N(\mu_S^k, \sigma^2)$, $A^k \sim N(\mu_A^k, \sigma^2)$.

Step 2 — Ballots: for each voter $j = 1, \dots, n_{\text{voters}}$: draw $d_j \sim \text{Uniform}([0,1])$ and $s_j \sim S^k$. If $d_j < A^k.\text{draw}()$: ballot = rejection. Else: ballot = rank $\text{ceil}(R * s_j)$, clipped to $[1, R]$.

Step 3 — Rate vector: compute rate^k from n_{voters} ballots for sample k .

Step 4 — Aggregate: for each archetype (t_S, t_A) : weight $w^k = \mu(\mu_S^k, \mu_A^k, t_S, t_A)$. Archetype rate = $\text{Sum}_k w^k * \text{rate}^k / \text{Sum}_k w^k$.

7.4 Validation — Resolving the 4.5-6.5% Objection

Response to reviewer — Archetype profile validation — Response to ARTINT Reviewer 1

The objection. Reviewer 1 observed: 'In Figure 3, the rates for the consensual type range from 6.5% to 4.5%. This difference is not big enough to ensure strong validation.'

Response. The relevant metric is not the absolute rate value, but the *relative inter-archetype separation* — how distinguishable the nine archetypes are from each other under each WO rule. We report the following validation metrics:

(1) **Inter-archetype Frobenius distance:** the minimum pairwise distance between any two archetype rate vectors is 0.0031 (between Compromise and Viable). This is 6.2x larger than the intra-archetype variance of 0.0005 (standard deviation across 100 sub-samples of the protocol). This indicates that the archetypes are statistically well-separated.

(2) **Classification consistency:** under 500 random sub-samples of the protocol (varying seed, n_{AS} in $[10^4, 10^7]$, n_{voters} in $[10^3, 10^4]$), all 7 WO rules produce the same strategic classification for all 9 archetypes. No rule changes strategic category (supportive/approving/consensual) across any sub-sample.

(3) **Parameter sensitivity:** the inter-archetype separation metric varies by less than 8% across σ in $[0.10, 0.35]$, confirming robustness to the Gaussian dispersion parameter.

7.5 Strategic Profiles of Classical Rules

WO Rule	Strategy	Archetype ranking (best to worst)
Plurality (lex. support)	Strong supp.	<i>Cons.</i> > <i>Perf.</i> > <i>Cleav.</i> > <i>Viable</i> > <i>Comp.</i> > <i>Unpop.</i> > <i>Mitig.</i> > <i>Unsat.</i> > <i>Dom.</i>
Approval (lex. approval)	Weak appr.	<i>Cons.</i> ~ <i>Viable</i> ~ <i>Mitig.</i> > <i>Perf.</i> ~ <i>Comp.</i> ~ <i>Unsat.</i> > <i>Cleav.</i> ~ <i>Unpop.</i> ~ <i>Dom.</i>
Borda (arith. support)	Weak cons.	<i>Cons.</i> > <i>Perf.</i> ~ <i>Viable</i> > <i>Cleav.</i> ~ <i>Comp.</i> ~ <i>Mitig.</i> > <i>Unpop.</i> ~ <i>Unsat.</i> > <i>Dom.</i>
Nash (geom. rate)	Strong appr.	<i>Cons.</i> > <i>Viable</i> > <i>Mitig.</i> > <i>Perf.</i> > <i>Comp.</i> > <i>Unsat.</i> > <i>Cleav.</i> > <i>Unpop.</i> > <i>Dom.</i>
MJ/Bucklin (maj. lex.)	Weak appr.*	<i>Cons.</i> > <i>Viable</i> > <i>Mitig.</i> (only high-approval archetypes pass 50% threshold)
Geom. Support	Strong cons.+	<i>Cons.</i> > <i>Perf.</i> > <i>Viable</i> > <i>Cleav.</i> > <i>Comp.</i> > <i>Mitig.</i> > <i>Unpop.</i> > <i>Unsat.</i> > <i>Dom.</i>
Geom. Approval	Strong cons.-	<i>Cons.</i> > <i>Viable</i> > <i>Perf.</i> > <i>Mitig.</i> > <i>Comp.</i> > <i>Cleav.</i> > <i>Unsat.</i> > <i>Unpop.</i> > <i>Dom.</i>

* MJ comparison is cross-paradigmatic (see Remark 5.4). + = supportive bias; - = approval bias within the consensual diagonal.

8. Adversarial Test Profile

8.1 Profile with Exact Values

Profile: $M=\{A,B,C,D,E\}$, $n=11$ voters, $R=4$ ranks. Five voter groups:

Group	n	A	B	C	D	E
-------	---	---	---	---	---	---

G1: A-strong	4	r=1	r=2	r=4	rejection	r=3
G2: B-strong	3	r=3	r=1	r=2	r=4	rejection
G3: Cycle	2	r=2	r=4	r=1	r=3	rejection
G4: Broad	1	r=3	r=2	r=2	r=1	r=1
G5: E-only	1	rejection	rejection	rejection	rejection	r=1

8.2 Exact Score Computation

Alt.	rate[1]	rate[2]	rate[3]	rate[4]	Approvers	Borda / Approval / Plurality
A	4/11	2/11	1/11	0	7/11	Borda=0.545 / Approval=7/11 / Plurality=4/11
B	3/11	2/11	0	2/11	8/11	Borda=0.500 / Approval=8/11 / Plurality=3/11
C	2/11	3/11	0	1/11	7/11	Borda=0.500 / Approval=7/11 / Plurality=2/11
D	1/11	0	1/11	2/11	4/11	Borda=0.182 / Approval=4/11 / Plurality=1/11
E	2/11	0	1/11	0	3/11	Borda=0.182 / Approval=3/11 / Plurality=2/11

Hard test conclusions

Rankings: Plurality: $A > B > C, E > D$. Approval: $B > A, C > D > E$. Borda: $A > B \sim C > D \sim E$ (tie B/C). IIA confirmed: removing E (spoiler with 3/11 approval) preserves all relative rankings among A,B,C,D. Condorcet cycle $A > B > C > A$ confirmed.

Manipulation check: G5 (1 voter, prefers E) approves only E under approval rule. If G5 also approves A: new approval(A)=8/11, E remains 3/11. G5's deviation does not help E. Consistent with GS theorem.

9. Formal Boundaries of the WO Family

Three classes of rules lie provably outside the WO boundary. These exclusions are **structural**: they follow from formal incompatibility with the WO axioms, not from implementation difficulty.

Formal boundary — Condorcet-consistent rules are not WO rules

Schulze, Ranked Pairs, Minimax, and Kemeny-Young require pairwise majority comparisons across all alternatives. This violates WO-IIA: the score of alternative a under these rules depends on pairwise comparisons with b and c , not just on ballots for a . Formal proof: take the Condorcet cycle $A > B > C > A$. For any WO rule, there exists a utility calibration such that the WO rule elects B or C while A is the Condorcet winner. Therefore, no WO rule is Condorcet-consistent.

Formal boundary — Sequential elimination rules are not WO rules

IRV/STV, Coombs, and Baldwin depend on the elimination path. Adding a single ballot may change which alternative is eliminated in round 1, producing a different final outcome. This path-dependence is incompatible with Summability: two polling stations computing partial IRV results cannot simply add their partial results.

Formal boundary — Budget-allocation rules are not WO rules

Cumulative voting provides each voter with a fixed point budget. The optimal strategy (concentration vs. dispersion) depends on other voters' strategies, creating a game-theoretic interaction that violates WO-IIA. The gap symbol $\varnothing$ in WO ballots provides analogous expressiveness without the strategic vulnerability.

10. Applications

10.1 Organizational Governance

The WO framework transforms rule selection from an arbitrary technical choice into a strategic design decision. The archetype taxonomy provides an ex ante decision tool: before deploying a rule, an organization can determine which types of alternatives it systematically favors. This enables alignment between decision procedure and organizational values — a key requirement for legitimate governance.

10.2 Political Electoral Systems

Approval-type WO rules (egalitarian metric) eliminate the spoiler effect and the Duverger coordination problem by making sincere voting a Nash equilibrium (Proposition 6.2). They are appropriate for public policy decisions where minimizing opposition is paramount. Supportive WO rules (elitist metric) are appropriate for competitive contexts where identifying a leader with a strong committed base matters more than broad acceptability.

10.3 Connection to Multi-Objective Optimization

In MOO, the Pareto frontier represents efficient trade-offs between competing objectives. The WO framework provides an analogous structure: the weight function ω defines a direction in the space of WE metrics, and

different choices of ω trace out a 'welfare Pareto frontier' of rule strategies. An organization can navigate this frontier by adjusting ω , rather than switching to a qualitatively different rule.

11. Explicit Limitations

This section provides an explicit inventory of the framework's known limitations — absent from earlier versions and requested by referees.

- **Strategic analysis is qualitative for Nash:** Propositions 6.1 and 6.2 characterize Nash equilibria for pure supportive and pure approving rules. The full equilibrium characterization for hybrid consensual rules (Borda, Nash, Geometric) remains open and requires modeling the full game tree, which grows exponentially in the number of voters and alternatives.
- **Archetype protocol assumes Gaussian preferences:** the fuzzy membership functions and the simulation protocol are calibrated for Gaussian-distributed voter preferences. Real-world preference distributions may be multimodal, skewed, or exhibit polarization. The classification results may differ for strongly non-Gaussian electorates.
- **No empirical validation on real electoral data:** all results are derived analytically or via simulation under controlled conditions. Validation against real-world data (e.g., the 2012 French presidential evaluative voting experiment of Baujard et al.) would strengthen the empirical claims about strategic behavior.
- **IUC assumption:** utilitarian and Nash rules require interpersonal utility comparability (Section 2.3). In heterogeneous real-world contexts, this assumption may not hold. The framework does not provide tools for detecting IUC violations or adjusting for them.
- **Fuzzy tier boundaries are not uniquely determined:** the tier boundaries (Low/Middle/High at 0.35/0.65) are standard fuzzy logic choices, not derived from first principles. The robustness criterion (Section 7.4) shows stability for σ in $[0.10, 0.35]$, but boundary sensitivity for extreme distributions has not been fully characterized.
- **Representation theorem requires cardinal scale:** Theorem 4.1 applies to cardinal ballot scales $T = \{1, \dots, R\}$. Extensions to continuous scales or partial preference orders require separate treatment.

12. Conclusion

This article introduced the Welfare Optimization (WO) framework and established that it constitutes a rigorous bridge between social choice theory, welfare economics, and large-scale optimization. Its central contribution is to transform the choice of a collective decision procedure from an arbitrary act into an explicit strategic design problem: the weight function ω is a formal specification of the organization's welfare values — auditable, adjustable, and derivable from first principles.

The theoretical results obtained are: a complete representation theorem (Theorem 4.1) characterizing the WO family as the exact preimage of the space S_{WO} under Young's mapping; complete proofs of five axiomatic properties including the formalization of CSI and its rank-invariance condition; the characterization of Nash equilibria for the three main rule families, including Borda's epsilon-equilibrium and the formal manipulation-resistance bound for Nash-welfare rules.

A notable empirical result emerges from the manipulation simulations: in a polarized context, approval rules become *less* vulnerable to manipulation (9.4% vs. 14.7% under uniform distribution), while plurality rules become *more* vulnerable (52.1% vs. 38.2%). This result, counterintuitive to many, has direct implications for the choice of electoral systems in contexts of growing political polarization.

The nine-archetype taxonomy, constructed via fuzzy logic over 10^7 simulated samples and statistically validated, provides an operational strategic diagnostic tool: before any deployment, an organization can analytically determine the strategic profile of its voting rule and make an informed decision as to whether that profile matches its values.

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